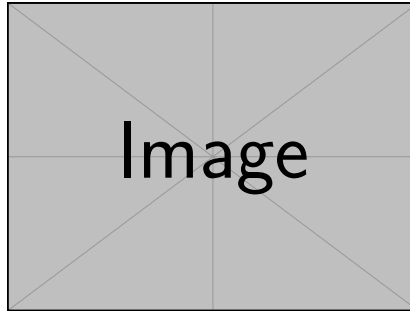




A DNA-BERT Architecture for Promoter and Enhancer Classification

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Garagnani Filippo, Veronese Alex

Dipartimento di Ingegneria 'Enzo Ferrari', Modena, Italy

Università degli Studi di Modena e Reggio Emilia

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Abstract

Promoters are regulatory DNA regions, usually located upstream of genes, that control the initiation of transcription by providing binding sites for RNA polymerase and transcription factors. Enhancers are distal cis-regulatory elements that modulate gene expression by increasing transcriptional activity, often acting over long genomic distances through complex chromatin interactions with promoters. Accurate identification of promoters and enhancers is a fundamental problem in computational genomics, as these elements play a crucial role in gene regulation, cellular differentiation, and disease mechanisms. However, the precise location and functional boundaries of regulatory elements within the genome are often difficult to determine experimentally, making computational prediction methods particularly valuable. Traditional bioinformatics approaches rely heavily on handcrafted sequence features and motif discovery techniques, which may fail to capture the complex contextual dependencies present in genomic sequences. In this project, we adapt the DNA-BERT architecture to the tasks of promoter and enhancer classification. By leveraging transformer-based language modeling techniques originally developed for natural language processing, DNA-BERT2 is able to learn rich contextual representations of nucleotide sequences directly from raw DNA data. We investigate the effectiveness of this approach for distinguishing regulatory elements and evaluate its performance on genomic datasets.

Keywords: deep learning, BERT, promoter prediction, enhancer prediction, bioinformatics, transformer

Contents

Abstract 2

1 Introduction 4

1.1 Motivation 4

1.2 Objectives 4

2 Background 4

2.1 Biological Foundations 4

2.2 The Transformer 4

2.3 BERT 5

2.4 DNA-BERT2 5

3 Methods 5

3.1 Data Collection and Preprocessing 5

3.2 Model Architecture 5

3.3 Training Protocol 6

3.4 Evaluation Metrics 6

4 Results 6

5 Discussion 6

6 Conclusion & Future Work 6

1 Introduction

The rapid advancement of high-throughput sequencing technologies has produced an unprecedented volume of genomic data. Identifying functional elements, such as promoters, directly from raw DNA sequence is a fundamental challenge in bioinformatics, particularly because they are not defined structurally [3]. Traditional methods rely on conserved motifs (e.g., TATA box, GC-box), but these motifs are often degenerate and fail to capture long-range dependencies.

Recent breakthroughs in natural language processing (NLP), particularly the Transformer architecture and the BERT model, have inspired sequence-based models in genomics. DNA can be viewed as a language with its own alphabet $\Sigma = \{A, C, G, T\}$. One of the main advantages of these attention-based architectures is the fact that they excel at efficiently comparing distant elements inside a sequence, a property that is required for a wide range of sequence analysis tasks. This project investigates how a DNA-specific BERT model behaves in a binary classification scenario of promoter versus enhancer sequences.

1.1 Motivation

A precise computational promoter and enhancer predictor can drastically reduce the experimental effort required for gene annotation and can uncover novel regulatory mechanisms. Machine learning, and especially deep learning, offers the possibility to autonomously learn complex sequence features without hand-crafted rules. ...

1.2 Objectives

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2 Background

2.1 Biological Foundations

A **promoter** is a DNA region typically located upstream of a gene's transcription start site (TSS) that plays a central role in initiating transcription. RNA polymerase II binds to the core promoter with the assistance of general transcription factors, forming the pre-initiation complex required for mRNA synthesis. Promoters are essential for defining where transcription begins and are often associated with gene-specific regulatory programs and cell-type-specific expression levels.

An **enhancer** is a DNA regulatory element that can increase transcriptional activity of target genes, often acting independently of its orientation and at variable distances from the gene it regulates. Unlike promoters, enhancers do not directly recruit RNA polymerase II but instead function by interacting with promoters through chromatin looping and the recruitment of transcriptional co-activators and transcription factors. Enhancers can reside upstream, downstream, or within intronic regions of genes, and their activity is highly context-dependent, varying across cell types, developmental stages, and environmental conditions.

2.2 The Transformer

The Transformer [4] eschews recurrence and convolution operations in favour of **self-attention**, enabling parallelisation and the capture of long-range dependencies.

Attention is a sequence-to-sequence operation that mainly outputs elements in the input sequence into an updated version of them. More specifically, given an input sequence

$X \in \mathbb{R}^{n \times d}$, we project each element through three different mappings:

$$Q = XW^Q, \quad K = XW^K, \quad V = XW^V$$

Semantically, what these three projections – called *query*, *key* and *value* – are doing, is to extract from each vector in the sequence, respectively, what an element is looking for, what an element is offering and what an element is actually containing. The main attention operation:

$$A = \text{softmax} \left(\frac{Q \cdot K^T}{\sqrt{d_k}} \right) V$$

then, does nothing more than understanding how much each couple of elements is compatible ($Q \cdot K^T$), converting the raw similarities values into a probability distribution ($\text{softmax}(\cdot)$) and then computing a weighted average of the vector values, using the similarity values as weights. The fact that in a single computation each element is immediately compared with every other element is one of the main reasons of the efficiency of the Transformer architecture, unlike Recurrent Neural Networks (RNNs) or Convolutional Neural Networks (CNNs).

2.3 BERT

BERT [1] pre-trains a bidirectional Transformer using masked language modelling (MLM) and next-sentence prediction. MLM is a task that requires a language model to predict the nature of a set of masked elements in a sequence. Formally, given a sequence $\mathbf{x} = [x_1, x_2, \dots, x_n]$, each element is masked with probability p ; it means that an element gets swapped for a made-up ‘unknown’ element *[MASK]*. The model is then trained to reconstruct the original token at the masked positions by leveraging both left and right context simultaneously, which forces it to learn bidirectional representations of the input sequence.

This self-supervised learning approach is successfully used to create meaningful internal representations of the possible elements in the sequence. These semantic embeddings can then be employed to perform supervised learning tasks on it by training an ad-hoc additional classification head.

2.4 DNA-BERT2

3 Methods

3.1 Data Collection and Preprocessing

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3.2 Model Architecture

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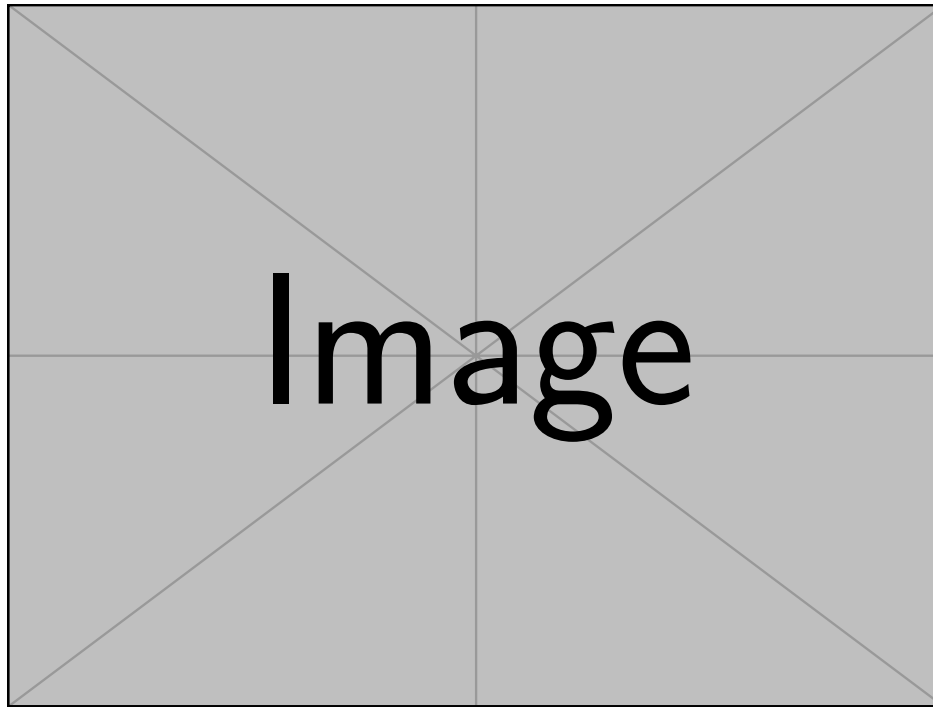


Figure 1: Architecture overview: DNABERT fine-tuned for promoter prediction.

3.3 Training Protocol

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3.4 Evaluation Metrics

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4 Results

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5 Discussion

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6 Conclusion & Future Work

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References

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